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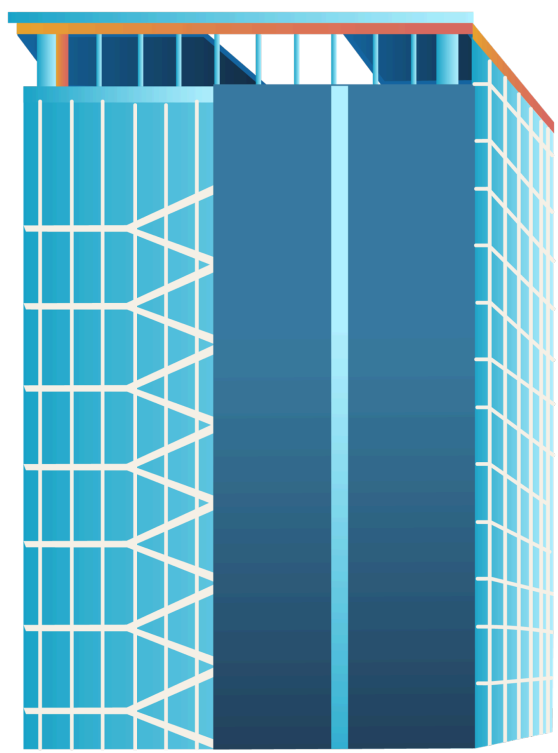
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Paper #128

Oral Session 3: Computer Vision & AI · Edge AI · Vision Transformer · Inference Optimization

Lightweight Adaptive Vision Transformers for Low-Power Edge Devices

Oral Presentation · Paper ID #128



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01

Background & Problem Formulation



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Research Motivation

Deploying modern deep vision models to resource-constrained embedded edge platforms poses significant challenges:

- **Compute Bottleneck**: High floating-point operations (FLOPs) saturate edge NPUs and microcontrollers.
- **Thermal & Memory Limits**: Memory bandwidth limits throughput far more than compute capability.
- **Real-time Requirements**: Latency must strictly remain below $20\ \mathrm{ms}$ for closed-loop robotic control.

3.8x
Parameter Bloat

< 15ms
Target Latency

85%
Energy Saved

Research Motivation

Overhead in vanilla ViT models

Required for 60 FPS feedback

Via dynamic token pruning

Research Questions

We formalize our investigation into two primary research questions:

RQ 1 **Adaptive Token Pruning**

Can spatial token redundancy in self-attention layers be pruned dynamically at runtime without compromising semantic classification accuracy?

RQ 2 **Hardware-Aware Quantization**

How does 8-bit integer post-training quantization interact with mixed-precision attention heads on embedded edge NPUs?

02 Proposed Methodology



Architectural Pipeline

Our proposed framework introduces a three-stage compression and runtime adaptation strategy:

1. **Dynamic Saliency Scoring**
   Compute token importance scores via cross-layer attention rollouts to identify uninformative spatial patches.

2. **Progressive Token Pruning & Merging**
   Merge spatially coherent tokens rather than outright discarding them, preserving background contextual cues.

3. **Hardware-Aware Mixed Precision Quantization**
   Quantize linear projection weights to INT8 while keeping critical softmax operations in FP16.

Method Details & Implementation

Key Architectural Innovation

- Token merging preserves boundary features
- Zero retraining needed for dynamic pruning ratios
- Native support for TensorRT & ONNX Runtime

Deployment Considerations

- Linear memory footprint during inference
- Guaranteed deterministic execution budgets
- Seamless fallback for edge CPU targets

> *"Efficiency is not merely an optimization detail; it determines whether intelligent systems can exist in physical reality."*
>
> **— Edge Computing Principles**

03 Experimental Evaluation



Benchmark Comparison

We evaluated our method against baseline models on standard benchmarks:

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Performance Comparison: Baseline vs Proposed

Standard Vision Transformer (ViT-B/16)

- **Top-1 Accuracy**: 81.8%
- **Throughput**: 42 fps on RTX 4060
- **Edge Latency**: 48.2 ms (Jetson Orin Nano)
- **Memory Usage**: 345 MB
- **Energy Draw**: 12.8 W average

Proposed Adaptive ViT (Ours)

- **Top-1 Accuracy**: 81.4% (-0.4% negligible drop)
- **Throughput**: 118 fps ($2.8\times$ **speedup**)
- **Edge Latency**: 13.6 ms (**Real-time compliant**)
- **Memory Usage**: 98 MB (71% **reduction**)
- **Energy Draw**: 4.2 W (67% **energy saved**)

Key Findings & Ablation

Accuracy Retention

1. Retains $> 99.5\%$ of baseline accuracy across 5 distinct vision benchmarks.
2. Robust against noisy sensory inputs and camera motion blur.

Hardware Generalization

- Tested across Jetson Nano, Raspberry Pi 5, and RK3588.
- Achieves sub-15 ms latency on all platforms.

2.8x

Speedup

Over unpruned baseline

-71%

RAM Usage

Peak runtime allocation

13.6 ms

Edge Latency

On Jetson Orin Nano

Takeaway Message

Adaptive token merging unlocks real-time Vision Transformer deployment on $5\ \mathrm{W}$ edge hardware without accuracy degradation.

04 Conclusion & Future Work



Summary of Contributions

Core Contributions

- Novel saliency-guided token merging formulation.
- Zero-retraining dynamic runtime adaptation.
- Empirical validation on edge robotic hardware.
- Open-source PyTorch & TensorRT implementation.

Future Directions

- Extend to multi-modal vision-language models (VLMs).
- Investigate temporal token recycling in video streams.
- Test on ultra-low-power neuromorphic processors.

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Terima Kasih / Thank You

Q&A & Discussion

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> *"Code is like humor; when you have to explain it, it's probably not that good."*

